$3^2 = 9$ $\quad 3^? = 9$

$3^x = 9 \iff \log_3 9 = 2$

$\log_a b = c \quad a^c = b$

1.

a) $\log_2 4 = 2$

b) $\log_3 27 = 3$

c) $\log_5 25 = 2$

d) $\log_{10} 100 = 2$

e) $\log_6 36 = 2$

2.

a) $\log_2 \frac{1}{2} = -1$

b) $\log_3 \frac{1}{9} = -2$

c) $\log_4 0{,}25 = -1$ ($0{,}25 = \frac{25}{100} = \frac{1}{4}$)

d) $\log_{10} 0{,}01 = -2$ ($0{,}01 = \frac{1}{100}$)

3.

a) $\log_3 \sqrt{3} = \frac{1}{2}$

b) $\log_5 \sqrt[3]{5} = \frac{1}{3}$

c) $\log_4 2 = \frac{1}{2}$

d) $\log_{81} 3 = \frac{1}{4}$

$4 = 2^2 \quad (2^2)^x = 2^1 \quad 2x = 1 \quad x = \frac{1}{2}$

4.

a) $\log_x 25 = 2$ $\quad x^2 = 25 \quad x^2 = 5^2 \quad x = 5$

b) $\log_5 x = 0$ $\quad 5^0 = x \quad 1 = x$

a) $\log_2 8 + \log_2 2 = \log_2 (8 \cdot 2) = \log_2 16 = 4$

b) $\log_3 9 - \log_3 3 = \log_3 \left(\frac{9}{3}\right) = \log_3 3 = 1$

a) $\log_6 4 + \log_6 9 = \log_6 (4 \cdot 9) = \log_6 36 = 2$

b) $\log_{\sqrt{2}} 2 = 2$

a) $\log_2^2 x = 4 \quad /\sqrt{\ }$

$\log_2 x = 2$

$2^2 = x$

$4 = x$

b)

$\log_a a^a = 1$ $\quad \log_a 1 = 0$ $\quad \log_a (b^m) = m \cdot \log_a b$

$\log_a b = \frac{\log_c b}{\log_c a}$ $\quad \frac{1}{\log_a b} = \log_b a$

$\log_{a^k} (b^m) = m \cdot \log_a b$

$\log_{a^k} b = \frac{1}{k} \log_a b$

$\log_{a^k} b = (\log_a b)^2$

$\log_{a^k} b^x = \log_a (b^x)$

c) $\log_{10} 250 + \log_{10} \frac{2}{5} = \log (250 \cdot \frac{2}{5}) = \log_{10} 100 = 2$

e) $\log_{10} 1250 - \log_{10} \frac{5}{4} = \log_{10} \left(\frac{1250}{\frac{5}{4}}\right) = \log_{10} (1250 \cdot \frac{4}{5}) = \log_{10} 1000 = 3$

a) $\log_2 1 + \frac{1}{2} \log_2 16 = 0 + 2 = 2$

$\log_2 (16^{\frac{1}{2}}) = \log_2 4 = 2$

e) $\log_{\frac{1}{16}} b = -\frac{3}{4}$

$\left(\frac{1}{16}\right)^{\left(-\frac{3}{4}\right)} = b$

$16^{\frac{3}{4}} = b$

$\left(\sqrt[4]{16}\right)^3 = b$

$2^3 = b$

$8 = b$